

PackageManifestInstaller

Dev Quick-Start Guide

v2.0 schema

Unity 6000.3+

URP

INVELON Technologies

What is this?

PackageManifestInstaller is a Unity Editor window that installs all required UPM packages for a project template in one click — no manual **manifest.json** editing needed.

It auto-discovers every ***.dependencies.json** file in the project (one per XLINK'd template) and shows each as a tab, listing required and optional packages with their install status.

Opening the window

From the Unity menu bar:

```
INVELON > Package Manager > Dependency Installer
```

Some templates also register a shortcut via the **menuGroup** field — e.g.:

```
INVELON > VR Template
```

Installing packages

Install single	Click Install on any row showing Missing or Low status.
Install All	Queues all pending required packages. Asset Store packages are always excluded — confirm them with Mark OK instead.
Refresh	Re-scans all installed versions. Also re-checks assetFolderPath presence for Asset Store packages.
Progress bar	Shown while installing. Don't close the window or switch scenes until it clears.

Package states

✓ OK	Installed version meets or exceeds the required minimum.
■ Low	Installed but below the required version — hit Update.
✗ Missing	Not found in the project. Hit Install (or follow the manual note for Asset Store).
... Loading	UPM query in progress.
✗ Error	UPM Client error. Check the Unity Console for details.

Package source types

	Source	Ver.	Reg.	Notes
REG	registry Unity Registry	✓	—	Standard UPM install. Uses id@version.

GIT	git Git repository	—	—	Full URL → Client.Add(). Supports #tag / #branch.
UPM	openupm OpenUPM	✓	✓	Installer auto-configures scopedRegistries.
TGZ	tarball Local .tgz file	✓	—	Drop the .tgz in Packages/ first.
AST	assetstore Unity Asset Store	—	—	Auto-detected via assetFolderPath if set, or confirmed manually with Mark OK button.

■ For Asset Store packages the Install button is replaced by a greyed-out Manual button. Follow the orange note under the row — it tells you which Unity account holds the licence and what to download from Package Manager > My Assets.

Adding a new template

1. Create **my-template.dependencies.json** anywhere inside the XLINK'd folder.
2. Confirm Unity indexes it as a TextAsset (must live inside **Assets/**).
3. Fill it following the schema below (or copy the example in the next section).
4. Hit **Refresh** — a new tab appears automatically, no code changes needed.

Minimal valid file:

```
{
  "schemaVersion": "2.0",
  "templateName": "My Template",
  "packages": [
    {
      "id": "com.unity.timeline",
      "version": "1.8.7",
      "source": "registry",
      "displayName": "Timeline"
    }
  ]
}
```

Schema reference (v2.0)

Root fields

Field	Type		Description
<code>schemaVersion</code>	string	✓ req	Always "2.0". Any other value blocks the tab from loading.
<code>templateName</code>	string	✓ req	Label shown on the tab in the installer.
<code>packages</code>	array	✓ req	List of package entries — see below.
<code>unityVersion</code>	string	opt	Minimum Unity version. Informational only. E.g. "6000.3".
<code>renderPipeline</code>	string	opt	Pipeline hint. Informational only. E.g. "URP".
<code>menuGroup</code>	string	opt	Registers an extra shortcut. E.g. "INVELON/VR Template".

Package entry fields

Field	Type		Description
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id	string	✓ req	UPM package identifier. E.g. "com.unity.xr.openxr".
version	string	✓ req	Minimum required version (semver). Informational for git and assetstore.
source	string	✓ req	"registry" "git" "openupm" "tarball" "assetstore"
displayName	string	✓ req	Human-readable name shown in the table.
url	string	◆ cond	Required for git (full URL + optional #tag) and openupm (https://package.openupm.com).
tgzFileName	string	◆ cond	Required for tarball. Filename of the .tgz in Packages/. Default: "{id}-{version}.tgz".
assetStoreId	string	opt	Numeric Asset Store product ID. E.g. "32416" for DOTween Pro.
scopeName	string	opt	openupm only. Registry display name in manifest.json.
optional	bool	opt	Moves the entry to the Optional section and excludes it from Install All. Default: false.
installNote	string	opt	Extra instruction shown under the row. Required by convention for assetstore. Format: "Login to [account] where you have the asset and download [name] [version] from Package Manager - My Assets."

✓ req = always required ◆ cond = required depending on source opt = optional

Example — VRCultureCore template

Shows all five source types, including one required and one optional Asset Store package:

```
{
  "schemaVersion": "2.0",
  "templateName": "VRCultureCore VR Template",
  "unityVersion": "6000.3",
  "renderPipeline": "URP",
  "menuGroup": "INVELON/VR Template",
  "packages": [
    { "id": "com.unity.xr.openxr", "version": "1.16.1", "source": "registry", "displayNa
me": "OpenXR Plugin" },
    { "id": "com.eflatun.sceneReference", "version": "5.0.0", "source": "openupm", "url": "ht
tps://package.openupm.com", "displayName": "Eflatun.SceneReference" },
    { "id": "com.xoia.basicscripts", "version": "0.0.22", "source": "tarball", "tgzFileN
ame": "com.xoia.basicscripts-0.0.22.tgz", "displayName": "XOIA Basic Scripts" },
    {
      "id": "com.demigiant.dotweenpro", "version": "1.0.410", "source": "assetstore", "assetSto
reId": "32416",
      "displayName": "DOTween Pro",
      "installNote": "Login to the Unity account where you have the asset and download DOTwee
n Pro 1.0.410 from Package Manager - My Assets."
    },
    {
      "id": "com.inter-illusion.i2localization", "version": "2.8", "source": "assetstore",
      "assetStoreId": "16831", "assetFolderPath": "Plugins/I2", "displayName": "I2 Localization
", "optional": true,
      "installNote": "Login to the Unity account where you have the asset and download I2 Loc
alization 2.8 from Package Manager - My Assets. Place under Assets/Plugins/I2."
    }
  ]
}
```

Exporting a snapshot

Click **Export current state** → **JSON** in the footer to write an **exported-YYYY-MM-DD-HHmm.dependencies.json** next to the active manifest. Handy for locking down exact versions or bootstrapping a new template from a working project.

■ *Asset Store packages are not tracked by UPM so they won't appear in the snapshot. Add them manually.*

Troubleshooting

No *.dependencies.json found	XLINK not applied or Unity hasn't indexed the file. Run Assets > Refresh.
Schema version error	schemaVersion is not "2.0". Update the field in your .dependencies.json.
JSON parse error	Syntax error in the file. Paste it into jsonlint.com to locate the issue.
Git package stays Missing	Wrong URL or private repo needs SSH. Try Package Manager > Add from git URL for the full error message.
Scoped registry not created	Non-standard manifest.json. Add the scopedRegistries block manually and retry.
Asset Store package always Missing	Expected until downloaded. Follow the orange row note → import via My Assets → hit Refresh.
Export generates empty file	UPM ListRequest failed. Check the Console and confirm network access.